

possible for the resultant electric field to be zero.

25 Ans: D

The electric force is pointing upwards and is balanced by the gravitational force which is pointing downwards.

Since the electric force is pointing upwards, the electric field line is pointing towards the upper plate which is at a lower potential. Hence, option A is wrong.

The electric field strength between the plates is uniform. Hence, option B is wrong.

When V decreases, the electric field strength decreases, and hence the electric force decreases. Particle P will fall towards the lower plate and as a result, gains electric potential energy and loses gravitational potential energy. Hence, option C is wrong.

When V increases, the electric field strength increases, and hence the electric force increases. Particle P will rise towards the upper plate and as a result, loses electric potential energy and gains gravitational potential energy.

26 Ans: A